
Tennis Analytics: Public Data, Prior Art, and Project Roadmap

Build the first version around a fixed panel of ten ATP and WTA events from 2021 through 2025, then test whether a transparent Four Factors model improves calibrated match forecasts beyond surface-adjusted Elo. This panel supplies 9,828 matches, with complete serve-point statistics for 9,755 of them (99.26%). It covers both tours, every surface, all four majors, and the six largest shared tour stops. The same design can later expand to the full tours and ATP Challengers.

This report audits the major free public data sources, official analytics systems, foundational research, and representative open projects that matter to that question. It distinguishes free access from legal reuse, and match-level statistics from point, shot, and tracking data. The audit used live source pages and repository snapshots on 14 September 2026. Public tennis data changes often, so every production pipeline should pin source versions and retain a provenance manifest.

Recommended research question

Can an interpretable Four Factors model capture tennis player strengths and produce better-calibrated match forecasts than surface-adjusted Elo?

What the evidence supports

- A credible Four Factors study is feasible now with free match-level data for ATP and WTA main tours.
- The strongest comparable 2021-2025 panel is the ten-event shared schedule described above.
- ATP Challenger expansion is feasible. Historical WTA ITF and men's Futures factor models are not consistent enough without another source.
- True optical tracking exists inside ATP, WTA, Wimbledon, US Open, and Roland-Garros systems, but no complete free public feed was found.
- IBM released useful historical Wimbledon and Grand Slam files. Its current AI features do not expose the underlying tracking feed.
- Elo remains the baseline to beat. The project succeeds even if Four Factors only improves explanation or calibration for selected surfaces or tours.

Bottom line

Start with one defensible project, not a data grab. Freeze the event panel, define a point-based net rating, estimate factor importance with out-of-sample and opponent-adjusted methods, then compare Four Factors, surface Elo, and their combination on a held-out season. Add point-by-point or video-derived features only after the simple model earns that complexity.

1. Scope and research method

No finite search can prove that it found every public tennis project. This audit instead targeted the public work most likely to affect the design: official ATP, WTA, ITF, Slam, IBM, Infosys, and Stats Perform systems; the principal open historical datasets; academic prediction and tracking papers; and representative current open-source implementations. Search results were checked against live pages, repository contents, licenses, and file-level coverage where downloads were available.

Audit tests

Test	Question asked	Why it matters
Access	Can a researcher obtain it without payment?	Free access alone does not grant copying, publication, or redistribution rights.
Provenance	Who created or preserved it, and from what source?	Mirrors and scraped copies can outlive the original but weaken the chain of custody.
Coverage	Which tours, levels, events, years, and grains appear?	A model cannot compare tours or seasons if the underlying competition changes unnoticed.
Completeness	Which fields are actually populated by event and year?	A column existing in a file does not make it analytically available.
Legitimacy	Do the terms permit research and derivative work?	A public GitHub or download link can still prohibit the intended project.
Model value	Can the source support explanatory factors, forecasts, or validation?	Each source should earn a specific role in the design.

Exact row counts in this report come from a local audit of pinned repository snapshots. The Jeff Sackmann originals were unavailable at their former GitHub addresses during the audit, so the counts use the Aneeshers archival mirror at commit 8373358. The mirror says it preserves June 2026 ATP/WTA snapshots and the October 2024 Slam point-by-point snapshot. Treat it as a preservation source and record checksums before analysis. [1]

Definitions

- Match level: one row per match, often with aggregate serve and return totals.
- Point level: one row per point or score state, which supports leverage, streak, and pressure analysis.
- Shot level: serve location, shot sequence, rally length, winners, errors, or net play.
- Tracking: player and ball coordinates over time. This is the richest layer and the least publicly available.

2. Free public data landscape

The useful ecosystem forms a ladder. Match-level archives provide the broadest historical coverage. Point and shot data add tactical detail for a smaller, uneven sample. Optical tracking drives the newest official metrics, but the raw coordinates remain proprietary.

Source	Grain / coverage	Free status	Best role	Decision
Sackmann archive mirror [1]	ATP 1968-2026; WTA 1977-2026; lower levels; match stats	CC BY-NC-SA 4.0; mirror of now-unavailable originals	Core match history, ratings, Four Factors	Use with pinned commit, attribution, noncommercial limits
Slam point-by-point [1]	1.92m points; 49 event-years, 2011-2024	Same license; feature coverage varies	Pressure, rally, serve direction, distance validation	Use only after feature-by-year audit
Match Charting Project mirror [2]	8,855 charted matches; 1.42m points; rich shot strings	CC BY-NC-SA; crowdsourced and selected	Tactical validation and prototypes	Use as supplementary, nonrepresentative sample
IBM Wimbledon datasets [3]	Grand Slam set data 2005-2015; Wimbledon PBP 2014-15	Apache 2.0	Historical Four Factors, serve speed, winners/errors/net play	Use; strongest permissive historical release
Live Tennis API academic [4]	173,571-match index, 2023-2026; point states across all tiers	Free on application; no raw redistribution	Lower-tier point-score research	Apply; validate provenance and partial tapes
tennis-data.co.uk [5]	ATP 2000-2026; WTA 2007-2026; odds	Free download; no clear open data license found	Market benchmark and calibration	Use locally only, seek permission before publishing derivatives
TennisData.app [6]	ATP/WTA/Challenger 2021-2026; odds	Download page conflicts with restrictive terms	Potential match and market feed	Do not use without written permission
Open Tennis Data v3 [7]	Top-level main draws from 2020; Parquet	Preview; source-specific obligations	Identity, provenance, date cross-check	Use as validator, not primary factor source
Broadcast-video CV sets [8-11]	Small labeled videos or derived coordinates	Mixed code/data/media licenses	Computer-vision experimentation	Use only with rights-cleared video

Decision rule: a source can be free and still be unsuitable for a public portfolio repository. Keep raw restricted data outside Git, publish download scripts and hashes where terms allow, and release only derived aggregates that the license permits.

3. Match-level coverage: what is actually usable

The archived Sackmann files are the practical backbone. Tour-level rows include tournament, surface, date, round, player identifiers, score, rankings, and winner/loser serve totals. ATP match statistics are generally present from 1991, Challenger statistics from 2008, and tour qualifying statistics from 2011. The WTA archive reaches 1977, but historical category codes require more normalization. [1]

Main-tour completeness, 2021-2025

Tour	Matches	Events	Both players have service-point totals	Rate
ATP	14,656	704 event-years	13,897	94.82%
WTA	13,485	520 event-years	12,900	95.66%
Combined	28,141	1,224 event-years	26,797	95.22%

The completeness rate is strong enough for a modern study, but the event list changes from year to year. Pandemic scheduling, tournament relocation, renamed stops, draw expansion, and source changes create false trends unless the project freezes a comparison panel or explicitly models tournament composition.

Important coding limits

- ATP code A combines ATP 250 and ATP 500. Recover the category from an official season registry instead of inferring it from draw size or prize money.
- WTA historical level labels changed across source systems. Map tournament identity first, then attach the official category for that season.
- Montreal and Toronto alternate by tour and year. A canonical event family should retain city, tour, and edition while grouping both as Canada for panel selection.
- Use stable player IDs where available. Normalize names only for source linkage, and retain the source name for audit.

Recommended canonical tournament key

tour + season + event_family + host_city + start_date. Store source tournament IDs as aliases. A separate event registry should hold surface, indoor/outdoor status when verified, official level, draw size, country, and lineage notes.

4. The standardized 2021-2025 panel

A shared ten-event panel creates the cleanest first comparison between ATP and WTA. It includes the Australian Open, Roland-Garros, Wimbledon, US Open, Indian Wells, Miami, Madrid, Rome, Canada, and Cincinnati. Every event family appears on both tours in every season from 2021 through 2025.

Season	ATP matches	ATP complete	WTA matches	WTA complete	Combined complete rate
2021	910	903	926	913	98.91%
2022	918	911	926	917	99.02%
2023	998	993	998	992	99.45%
2024	998	991	998	994	99.45%
2025	1,078	1,074	1,078	1,067	99.30%
Total	4,902	4,872	4,926	4,883	99.26%

Why this panel works

- It balances three hard-court regions with clay, grass, and all four majors.
- It avoids category ambiguity inside ATP 250/500 and the changing WTA lower-level labels.
- It compares the tours on the same event families while preserving tour-specific models.
- It gives enough observations for rolling player metrics, temporal holdouts, and surface interactions.

What it does not solve

Draws expanded during the period, and the 2021 calendar still reflects pandemic effects. The panel favors elite fields, so its factor weights may not generalize to ATP 250, WTA 250, Challenger, or ITF competition. Report this as the elite-event model, then test transportability on the full tour.

Recommended split

Use 2021-2023 for development, 2024 for model and factor selection, and 2025 as a locked final test. Update ratings and rolling statistics in chronological order. Never let a later match contribute to an earlier player feature.

5. How far down the tournament ladder can the project go?

Match results extend far below the main tours. Consistent factor inputs do not. That distinction sets the responsible boundary. ATP Challengers support a serious extension; older Futures and WTA ITF files mainly support Elo or result-based work until a separate point source is secured.

Competition	2021-2025 audited volume	Aggregate-stat continuity	Recommended use
ATP tour main draws	14,656 matches	High, 94.82% complete	Core Four Factors and forecast study
WTA tour main draws	13,485 matches	High, 95.66% complete	Core Four Factors and forecast study
ATP Challenger plus tour qualifying	50,995 rows	Very high, 99%+ in each audited year	Second-stage Challenger model after event-type separation
Men's Futures / ITF	82,094 rows	0% in 2021-2024, then 97.3% in 2025	Historical Elo only; 2025 factors require source-change validation
WTA qualifying / ITF	151,512 rows	About 7% in 2021-2024, then 98.0% in 2025	Historical Elo only; do not pool pre/post source regimes
Live Tennis academic point data	173,571-match index across all tiers, 2023-Aug 2026	Point tapes available for a subset; full access by application	Prospective lower-tier extension after tape validation

The jump from almost no lower-tier aggregate statistics to near-complete coverage in 2025 looks like a source-regime change, not a sudden improvement in player measurement. Analyze the two regimes separately.

Official level map

- ATP: Grand Slam, Masters 1000, ATP 500, ATP 250, Challenger 175/125/100/75/50, then ITF World Tennis Tour M25/M15. [12-13]
- WTA: Grand Slam, WTA 1000, WTA 500, WTA 250, WTA 125, then ITF World Tennis Tour W100/W75/W50/W35/W15. ITF women's labels changed in 2024, so retain season-specific categories. [14-16]

Expansion rule

Add a tier only when at least 90% of matches in every included season contain the fields required by the frozen factor definitions. A lower threshold can be studied, but it should form a separate cohort with missingness modeled and reported.

6. Point, shot, and tracking data

Public point data can test how aggregate factors arise. It cannot yet supply a uniform tour-wide tracking layer. The main public files differ by tournament, season, and feature collection, so a coverage matrix must precede modeling.

Slam point-by-point archive

The audited snapshot contains 1,922,136 point rows across 49 singles event-years from 2011 through 2024. All four majors appear through 2019; 2020 excludes the canceled Wimbledon; all four appear in 2021; the local snapshot then contains Wimbledon and the US Open for 2022-2024. Fields include server, point winner, serve number and speed, rally sequence and count, serve direction/width/depth, return depth, winners and errors, net play, break-point state, and player distance run. [1]

Feature example	Observed pattern	Modeling consequence
Rally count	Roughly 94-95% nonzero at 2019 Wimbledon/US Open; 49-90% in sampled 2022-2024 files	Build event-year completeness gates; do not impute unrecorded rallies as zero
Serve speed	Roughly 48-92% populated depending on event/year	Standardize units and analyze only supported cohorts
Distance run	Present for many Wimbledon/US Open files, absent at AO/FO	Confirm whether values are point increments or cumulative states before use
Shot sequence	Rich encoded strings, but conventions vary	Parse with unit tests and retain raw encodings

Match Charting Project

The audited mirror contains 8,855 valid matches and about 1.42 million point rows through 18 October 2024. It records serve direction and type, rally shot strings, return depth, shot direction, errors, net approaches, and serve-and-volley. The sample is crowdsourced and nonrandom. Use it to validate mechanisms or prototype tactical metrics, not to estimate tour prevalence without selection adjustment. [2]

Academic live point feed

Live Tennis API advertises a free academic program with a 173,571-match index from 1 January 2023 through 13 August 2026 across ATP, WTA, Challenger, and ITF competition. The public Zenodo sample includes the full match index, player metadata, and a June 2026 point sample. The provider reports 170,527 matches with point sequences across its layers and 22,031 matches observed live since 12 October 2025. Earlier sequences may be reconstructed from a third-party feed. Tapes can begin or end mid-match, and raw redistribution is prohibited. [4,17]

7. IBM and the official AI analytics ecosystem

IBM does release historical tennis data, but the public release is small and old compared with the systems now used at Wimbledon and the US Open. The current products expose scores, narratives, forecasts, and selected metrics to fans. This audit found no public bulk endpoint for the underlying contemporary tracking feed.

What IBM released

File	Audited contents	Analytic value
Slams_Sets_201605310936.csv	86,934 player-set rows; 39 columns; all four majors, mostly 2005-2015 plus 2016 AO; both tours	Serve/return, winners, unforced errors, net play, break points, first/second serve speed. Excellent permissive historical factor laboratory.
Wimbledon 2014 and 2015 point files	83,307 point rows; rounds 1-3; men and women; 28 columns	Serve direction, speed, return hand, rally count, shot outcome, timestamps. Useful for point-level prototypes.

The repository uses Apache 2.0. That makes it easier to redistribute code and derived work than the noncommercial Sackmann license, subject to normal notice and attribution requirements. [3]

What IBM uses now

- Live Likelihood to Win updates match probability point by point.
- Power Index and Keys to the Match combine historical and current performance signals.
- Key Moments identifies pivotal passages for highlights and summaries.
- Serve Quality, introduced for the 2026 US Open, uses limb, racket, and ball tracking at 50 Hz across 21 tracked points to evaluate efficiency, accuracy, consistency, and toss. [18-21]

IBM reports that its US Open architecture combines long-run historical records, millions of real-time data points, rankings, head-to-head information, live statistics, media, and licensed Sportradar data. Those descriptions explain the product, not a reproducible public dataset. [20]

Other official systems

- ATP Tennis Data Innovations and TennisViz generate more than 60 shot types, 0-10 shot quality, attack/neutral/defense states, seven tactics, and 48 patterns. Tennis IQ gives players access; the raw data is proprietary. [22-25]
- Stats Perform holds official WTA data rights and supplies AI-enriched Opta feeds, including clutch-point and win-probability products. These feeds are licensed, not free. [26]
- Infosys Stats+ ranks statistics by disproportionate influence, combining volume and contextual importance. It is the closest commercial analogue to a match-specific factor-importance system, but its formula and inputs are not published. [27-29]

8. What analysts and researchers have already done

The project sits at the intersection of three mature lines of work: point-based player strength, Elo-style match forecasting, and tracking-based shot value. Its novelty should come from joining interpretation and calibration under a transparent, reproducible design.

Work	Method and finding	Lesson for this project
Pillitere, Analytics in the ATP [30]	Linear regression on ATP 1991-2014; compares serve and return components across eras	Closest published precedent to regression-derived factor importance. Extend to WTA, surfaces, calibration, temporal validation, and reproducible weights.
Kovalchik, GOAT of tennis prediction [31]	Compared 11 models on 2,395 ATP matches; rankings and Elo were strong; emphasized calibration and log loss	Elo is the benchmark. Accuracy alone is inadequate.
Sipko and Knottenbelt [32]	Compared ranking and Elo variants across ATP/WTa Slam samples; surface Elo helped in ATP tests	Estimate and report tour-specific behavior instead of assuming one formula transfers.
Gollub et al. [33]	Opponent-adjusted serve ratings with shrinkage and point-based match models	Adjust factor inputs for opponent and sample size before assigning weights.
Barnett and Clarke [34]	Combined serve/return statistics with tournament context, then used tennis scoring probabilities	A mechanistic point-to-match model is a useful alternative to direct logistic regression.
Network centrality [35]	Eigenvector centrality on 2016-2020 men's and women's result networks	Include one abstract strength baseline beyond Elo if time permits.
Expected Shot Value / VON CRAMM [36]	Tracking-based shot execution, selection, and court-coverage values at the 2019 US Open	Shows what true tracking can add, but the source coordinates are not broadly public.
Unified Server Quality [37]	Mixed-effects model using public Wimbledon/USO point data; serve efficiency within three shots, speed, location, entropy, server/returner effects	Best direct rich-data replication candidate after the Four Factors foundation.
O'Shannessy First Four Shots [38]	Popular tactical framework emphasizing points decided in the first four shots	A clear rally-length factor family, but it requires point/shot data and careful attribution.

This audit did not find a transparent, published ATP-and-WTA system that derives exactly four stable factor weights from a net tennis rating and then tests those factors against a calibrated surface-Elo forecast. That is a credible niche, provided the factor count is treated as a design constraint and not proof that tennis naturally has four independent dimensions.

9. Translating Dean Oliver's logic to tennis

Dean Oliver began with a team outcome measured per possession, selected conceptually distinct ways teams create that outcome, and estimated their unequal importance. Tennis needs the same separation between the outcome, factor families, and forecasting test.

Primary outcome: Net Point Rating

For player i over a defined window: $NPR = 100 \times (\text{points won} - \text{points lost}) / \text{total points}$. Because points won plus points lost equals total points, NPR also equals $200 \times \text{point-win percentage} - 100$. A value of +4 means the player won four more points than lost per 100 points played.

Equal-phase sensitivity outcome

Use $100 \times (\text{service points won percentage} + \text{return points won percentage} - 1)$. This gives service and return equal influence even when a player has faced unequal numbers of serve and return points. Compare its factor weights with NPR.

Secondary outcomes

- Net Game Rating: service games won minus return games lost per 100 games.
- Match win probability: the forecasting target, not the regression outcome used to discover factor importance.
- Surface-specific residual strength: performance above expectation after accounting for opponent and surface.

Candidate Four Factors

Factor family	Core match-level ingredients	Meaning	Main caution
1. Serve creation	Ace rate, first-serve-in rate, first-serve points won	Creates immediate advantage and protects short service points	First-in rate and first-serve success trade off; model jointly
2. Second-serve security	Second-serve points won, double-fault rate	Measures resilience after the first serve misses	Double faults are already embedded in some denominators
3. Return pressure	First- and second-serve return points won; break chances created per return game	Turns opponent service points into neutral or attacking positions	Raw break chances depend on opponent and score path
4. Conversion and recovery	Break conversion and break saving above model expectation	Captures execution on high-leverage points after controlling for baseline point skill	Raw break-point percentage is noisy and often mislabeled as clutch

These are hypotheses, not finished factors. The analysis should compare alternative definitions inside each family and freeze the winning specification before the final 2025 test.

10. How to derive weights without fooling ourselves

A multiple linear regression can estimate how player-season-surface factor values relate to Net Point Rating. Raw coefficients alone should not become the published weights. Units differ, predictors overlap, and repeated observations from the same player violate simple independence assumptions.

Recommended estimation sequence

1. Define one observation as a player over a rolling 52-week surface window, or as a player-season-surface. Require a minimum number of service and return points.
2. Construct every input from matches completed before the evaluation date. Apply empirical-Bayes or ridge-style shrinkage toward tour and surface averages for small samples.
3. Adjust component rates for opponent strength, surface, season, and tournament environment. A mixed model with player and opponent effects is a defensible second specification.
4. Standardize predictors within tour and surface before comparing coefficients.
5. Estimate ordinary regression for interpretability, then use LMG/Shapley decomposition or dominance analysis to allocate model R-squared across correlated factor families.
6. Bootstrap by player, not row, to quantify how often the factor ordering and weight shares remain stable.
7. Choose the factor definitions with 2021-2024 only. Lock formulas and weights, then score 2025 once.

Suggested model ladder

Model	Purpose	Minimum standard
Unadjusted linear model	Transparent starting point	Standardized factors; player-clustered uncertainty
Opponent-adjusted mixed model	Separates player skill from schedule	Server, returner, surface, season, and event context
Regularized sensitivity model	Tests coefficient stability under collinearity	Penalty tuned only inside development years
LMG/Shapley importance	Converts shared explained variance into weight shares	Bootstrap intervals and tour-specific estimates

What the final weights should mean

Publish weights as each factor family's share of the explained variance in out-of-sample Net Point Rating, accompanied by uncertainty and stability across ATP, WTA, and surfaces. Do not present them as universal laws if the confidence intervals overlap or the ordering changes.

11. The forecast test

The explanatory regression and the match forecast answer different questions. The first asks which skill families explain point dominance. The second asks whether the resulting player ratings predict unseen matches better than Elo.

Models to compare

Model	Inputs	What it tests
Ranking baseline	Pre-match official rank or rank points	A familiar public benchmark
Overall Elo	Chronological match results	Pure result-based dynamic strength
Surface-adjusted Elo	Overall and surface-specific Elo with shrinkage	The principal benchmark
Four Factors	Pre-match opponent-adjusted factor ratings	Whether interpretable components predict match outcomes
Elo + Four Factors	Surface Elo plus factor differences	Whether factors add information beyond results
Optional network score	Pre-match result-network centrality	Whether a more abstract strength representation adds value

Evaluation priorities

- Primary: Brier score, log loss, calibration intercept and slope, and reliability curves.
- Secondary: accuracy, ROC AUC, and upset recall. These do not replace calibration.
- Report uncertainty with a block bootstrap by tournament or week.
- Evaluate ATP, WTA, and each surface separately before pooling conclusions.

Leakage controls

- Build features in player A/player B orientation, then randomly swap slots in tests to verify symmetric probabilities.
- Never use winner/loser columns to construct a pre-match feature after the row has been oriented by outcome.
- Lag every rolling window and rating update until after the match prediction is recorded.
- Keep retirement and walkover rules explicit. Score sensitivity analyses with and without retirements.
- Freeze preprocessing, factor definitions, and hyperparameters before opening the 2025 test results

A Four Factors model that matches Elo's Brier score but explains why players differ is still valuable. A combined model that improves calibration without improving raw accuracy is also a legitimate finding.

12. Project portfolio: ranked ideas

The first six ideas form one coherent roadmap. The remaining ideas are optional branches that reuse the same cleaned data, player identities, and evaluation framework.

#	Project	Data	Core method	Fit
1	ATP/WTa Four Factors	2021-2025 ten-event panel	Opponent-adjusted regression plus Shapley/LMG importance	Best starting project
2	Four Factors vs surface Elo	Same panel	Chronological probabilistic forecast comparison	Directly answers the research question
3	Elo + Factors hybrid	Same panel	Logistic model using Elo and factor differences	Tests incremental value
4	Tour transfer	ATP and WTA panel	Estimate weights separately; cross-apply and test	Shows whether one tennis model fits both tours
5	Surface portability	Hard, clay, grass panel	Hierarchical interactions and stability intervals	Finds which strengths travel
6	Full-tour validation	All ATP/WTa 2021-2025	Apply frozen elite-event factors	Tests external validity
7	Challenger readiness	ATP Challenger stats plus main tour	Tier-adjusted factors and promotion forecasting	Strong second-stage extension
8	IBM Grand Slam Four Factors	IBM sets 2005-2015	Serve, return, errors/winners, net play, speed	Rich permissive historical study

12. Project portfolio: tactical and advanced branches

#	Project	Data	Core method	Fit
9	Public Serve Effectiveness	Slam PBP / MCP	Unreturned serve or server advantage within first ball	Transparent proxy for TennisViz metric
10	Unified Server Quality replication	Wimbledon/USO PBP	Mixed-effects serve-efficiency model	Closest rigorous tracking-adjacent replication
11	Return Pressure rating	Match stats plus selected PBP	Opponent-adjusted return points and break creation	Could become the strongest factor family
12	Clutch residual	Point scores or break-point aggregates	Actual pressure-point wins minus baseline expectation	Separates leverage from raw skill
13	Attack / Conversion / Steal proxies	Shot-charted points	Classify point state from shot strings and rally outcomes	Public analogue to ATP Insights
14	Rally style fingerprints	MCP / Slam PBP	Rally-length and direction mixtures	Tactical player profiles, not first forecast target
15	Tournament environment ratings	Match aggregates	Estimate court-speed and scoring environment effects	Improves opponent and event adjustment
16	Lower-tier point model	Live Tennis academic	Point-state Elo and factor estimation across tiers	High upside after access and tape audit
17	Network strength baseline	Match results	Time-decayed eigenvector centrality	Abstract alternative requested by the user
18	Video movement pressure	Rights-cleared match video	TrackNet/OpenCV court coordinates and time-to-contact	Separate Python/CV project; high effort

Recommended sequence

Complete projects 1-3 as one portfolio analysis. Add 4-6 if the result is stable. Choose either 7 for broader competition coverage or 8-10 for richer point and shot interpretation. Treat 16 and 18 as separate future projects because access, media rights, and engineering effort change the scope.

13. Languages, software, and reproducible workflow

Use R for the main project. Its strengths match the work: table cleaning, grouped rate construction, statistical modeling, calibration graphics, and a reproducible report. Python becomes necessary only for computer vision or a specialized tracking pipeline.

Layer	Recommended tools	Reason
Data cleaning	R: readr, dplyr, tidyr, stringr, lubridate	Readable tidyverse workflow and strong grouped transformations
Exploration and plots	R: ggplot2	Consistent publication graphics and easy export to SVG for the portfolio
Models	Base R lm/glm; lme4; boot; relaimpo or a tested custom LMG routine	Starts simple, then adds opponent effects, uncertainty, and correlated-predictor importance
Prediction scoring	yardstick or small audited functions	Brier, log loss, calibration, ROC AUC, and resampling
Pipeline	targets	Rebuilds only the stages affected by a data or code change
Environment	renv	Pins R package versions so the analysis remains reproducible
Reporting	Quarto	Combines prose, tables, methods, and outputs in one reproducible document
Large point files	Arrow or DuckDB only if needed	Avoids loading millions of rows into memory; unnecessary for the first match-level version
Computer vision	Python: OpenCV, PyTorch, TrackNet-style models	Required only for video-derived coordinates and pose
Version control	Git and GitHub; no raw restricted data	Keeps code public without violating source terms or bloating history

Suggested repository boundaries

- README.md: research question, current status, and how to reproduce.
- DATA_LICENSE.md: source-by-source rights, attribution, and redistribution decisions.
- data-raw/: gitignored source files; download manifest with URL, access date, commit, hash, and terms.
- data-derived/: only aggregates or analysis-ready files the source license permits.
- R/: small functions for ingestion, identity linkage, metrics, Elo, models, and scoring.
- analysis/: Quarto notebooks that call tested functions instead of containing the whole pipeline.
- outputs/: tables, model cards, and SVG plots for later website integration.

14. Standardization contract

A single canonical match table should isolate source quirks from analysis logic. Raw files remain untouched; transformations create versioned, testable tables.

Table	Required fields	Key checks
events	event_id, family, tour, season, city, country, level, surface, dates, draw	One edition per tour-season-family; official level source retained
players	player_id, source IDs, canonical name, hand, birth date when permitted	No name-only joins after linkage; ambiguous matches quarantined
matches	match_id, event_id, date, round, best_of, player_a/b, score, status	Player slots are outcome-neutral; walkovers excluded
match_stats	serve points, first serves, aces, double faults, first/second won, break points, service games	Nonnegative; component counts cannot exceed denominators; both players reconciled
ratings_pre	match_id, player, overall Elo, surface Elo, factor ratings, windows	Every value timestamped before match start
predictions	match_id, model_version, p_a, outcome, split	$0 \leq p \leq 1$; swap symmetry; frozen test versions
source_manifest	URL, license/terms URL, accessed_at, commit/version, SHA-256, transformations	One record for every raw file; no silent replacements

Tournament identity rules

- Use official calendars to define level and event identity; use scraped archives only for match rows.
- Keep event family separate from host city so alternating and relocated tournaments can be compared without erasing history.
- Store both the source category and the canonical category. Never overwrite the evidence.
- Version the panel membership. A tournament enters the five-year panel only if the same event family appears for both tours in each included year.

Field availability contract

For each variable, publish a matrix of nonmissing rate by source, tour, event, and year. A zero can mean a real zero only when the source documentation and count constraints support that interpretation. Missing features remain missing.

15. Practical build plan

Phase 1: prove the data contract

1. Pin the archive commit, store file hashes, record the license, and create the ten-event registry.
2. Parse 2021-2025 ATP and WTA matches into outcome-neutral player A/player B rows.
3. Produce coverage, count, denominator, and identity reports. Stop if any event-season fails the contract.

Phase 2: establish baselines

4. Implement overall Elo and surface Elo with chronological updates.
5. Create rank and Elo predictions, then lock the 2025 scoring harness.
6. Confirm swap symmetry, temporal leakage tests, and retirement rules.

Phase 3: derive Four Factors

7. Construct candidate serve creation, second-serve security, return pressure, and conversion/recovery measures.
8. Fit tour-specific opponent-adjusted models and estimate factor importance with player bootstrap intervals.
9. Freeze the four definitions and weights using development and validation years only.

Phase 4: answer the question

10. Generate untouched 2025 predictions for rank, Elo, Four Factors, and the hybrid.
11. Compare Brier score, log loss, calibration, and subgroup results by tour and surface.
12. Write the result even if Four Factors does not beat Elo. Separate explanatory value from forecasting value.

Go / no-go criteria

Gate	Go	No-go response
Panel coverage	$\geq 95\%$ of matches have required counts in every tour-year	Narrow the factor definitions or event cohort
Weight stability	Ordering and shares remain broadly stable in player bootstrap	Report factors without universal weights
Calibration	Four Factors or hybrid improves held-out Brier/log loss or calibration	Keep Elo as forecast model; present factors as explanatory ratings
Tier transfer	Frozen model retains calibration outside elite panel	Estimate a tier-specific model and label the scope

16. Risks, decisions, and final recommendation

Principal risks

- License drift: repositories and download pages change. Preserve the exact terms and access date in the manifest.
- Selection bias: charted or tracked matches overrepresent stars, majors, and televised courts.
- Survivorship: using current player metadata or final ranks can leak future information.
- Collinearity: serve and return ingredients overlap mechanically with point-based outcomes.
- Context: raw break-point rates and winner/error counts reflect opponents, surfaces, score paths, and recording conventions.
- False comparability: category names, cities, surfaces, and data providers can change while the tournament label looks stable.

Decisions to make before coding

1. Confirm the project is noncommercial while it uses CC BY-NC-SA data.
2. Choose whether the first release covers the ten-event panel only or uses that panel for discovery and the full tour for validation.
3. Choose the rolling window and minimum point threshold before inspecting final test performance.
4. Decide whether conversion/recovery must be a factor. If its residual signal is unstable, replace it with a less score-dependent family such as rally tolerance when point data permits.

Final recommendation

Build one integrated project with three outputs: an interpretable ATP/WTa Four Factors rating, a surface-Elo benchmark, and a calibrated hybrid forecast. Use the ten shared events from 2021-2025 as the first clean laboratory. Treat ATP and WTA as parallel models, not one pooled population. Publish the data contract, factor uncertainty, and negative results. After that foundation is stable, extend either downward to Challengers or inward to point-level serve and return quality.

This path answers the original question with free data, creates a credible statistical narrative for a portfolio, and leaves room for richer tracking-style metrics without making proprietary coordinates a prerequisite.

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Audit note

Coverage counts are direct file audits, not claims made by the listed sources. Official product descriptions were used only to describe announced metrics and data flows. Where no public raw feed or clear open license was found, the report says so rather than treating absence of evidence as a grant of permission.